# Office Hours 10

1. Consider the system shown in Fig. 1. A block slides down a frictionless ramp under the force of gravity. At the same time, the ramp is being rotated with an angular speed $\omega$. Find the equation(s) of motion.

2. Consider two equal masses connected by a string that goes through a hole in a table, as shown in Fig. 2. The mass on top of the table slides on the table without friction. Find the equations of motion for this system. Can you find a stable equilibrium?

3. Consider a block sliding down a frictionless wedge as shown in Fig. 3. The wedge slides on a frictionless surface. Write down the Lagrangian and equations of motion for:

   (a) The block-wedge system shown in Fig. 3a

   (b) The block-wedge-spring system shown in Fig. 3b

   (c) The block-wedge system with a pendulum attached to the block, as shown in Fig. 3c

   (d) The block-wedge system with a pendulum attached to the block. The pendulum has a spring instead of a string, as shown in Fig. 3d

   (e) The block-wedge-spring system with a pendulum attached to the block, as shown in Fig. 3e

   (f) The block-wedge system shown with a pendulum attached to the block. The pendulum has a second spring instead of a string, as shown in Fig. 3f

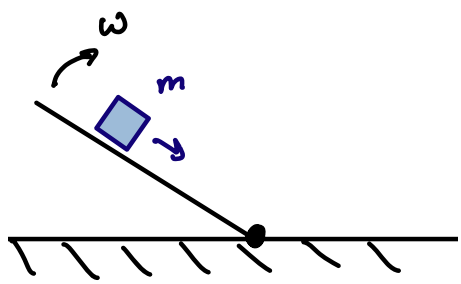


Figure 1: The block sliding down a rotating ramp

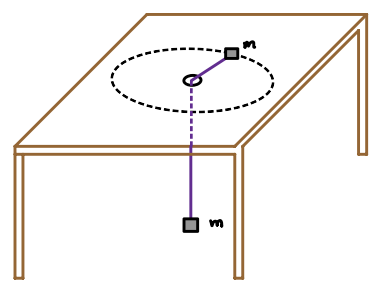


Figure 2: The two masses attached to a string through a hold in a table.

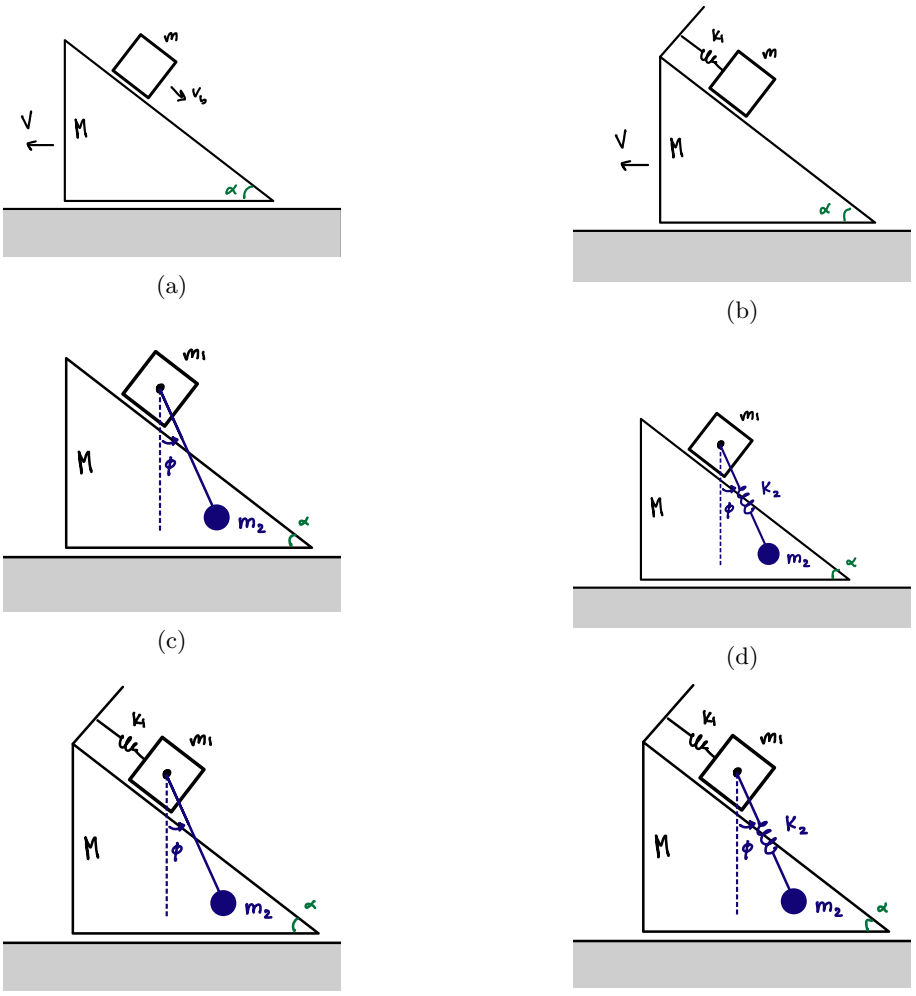
m
$v_b$
V
M
α
(a)
$k_1$
m
V
M
α
(b)
$m_1$
M
ϕ
$m_2$
α
(c)
$m_1$
M
$k_2$
ϕ
$m_2$
α
(d)
$k_1$
$m_1$
M
ϕ
$m_2$
α
(e)
$k_1$
$m_1$
M
$k_2$
ϕ
$m_2$
α
(f)

Figure 3